Genetic association, macrophage states and the limits of evidence in public datasets

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Open data · Reproducible workflow · Nature-style data figures

Exploratory RA integration with an NSCLC extension

The scores used in this study

S1 · 15 discovery genes	From sorted active/remission macrophages. Used for exploratory transfer and anti-TNF change.
S2 · 10 predefined IFN genes	Associations in spatial tissue and across bulk disease stages. Only 2 genes overlap with S1.
Hallmark IFN-gamma	Pathway and protein evidence. Specificity is supported in lining, not in sublining.
M-6 / Tph · 30 genes each	Programmes from atlas states, transferred to the fluid and lung cohorts.

Published atlas states and tested integration baselines

31,520 nuclei from 12 donors. RNA and ATAC are measured in the same nuclei.

Fine-state labels, including the 65-nucleus M-6, are taken from Gupta et al.

Concatenation and WNN rank highest in this benchmark. Rare M-6 neighbourhoods favour RNA alone.



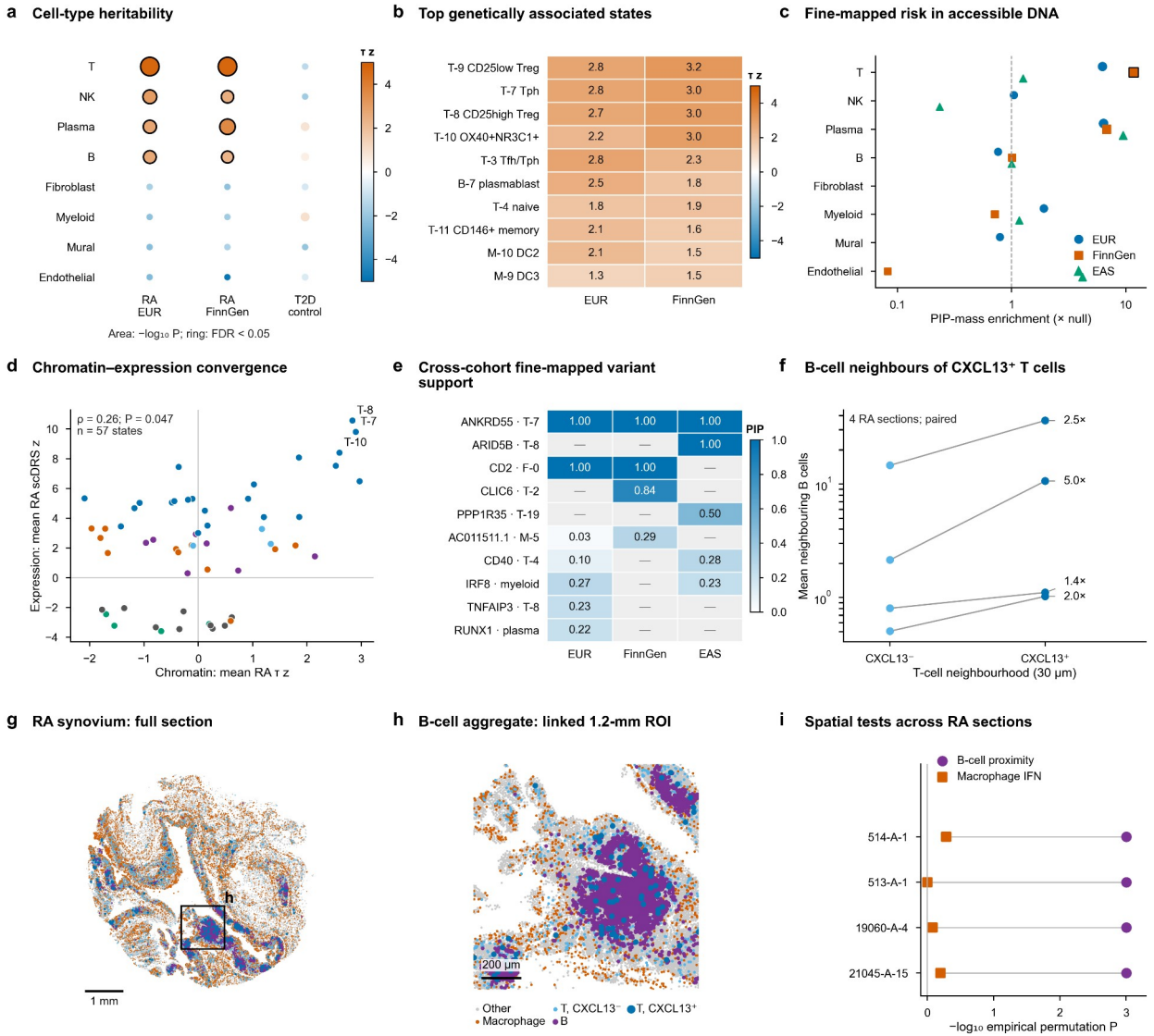
Genetic enrichment and spatial neighbourhood associations

Tph/Treg-associated chromatin is enriched for RA heritability.

S-LDSC and scDRS are complementary methods that use the same GWAS traits.

More B cells sit near CXCL13-positive T cells in 4/4 sections.

The empirical global-permutation P does not account for label clustering or tissue compartments.



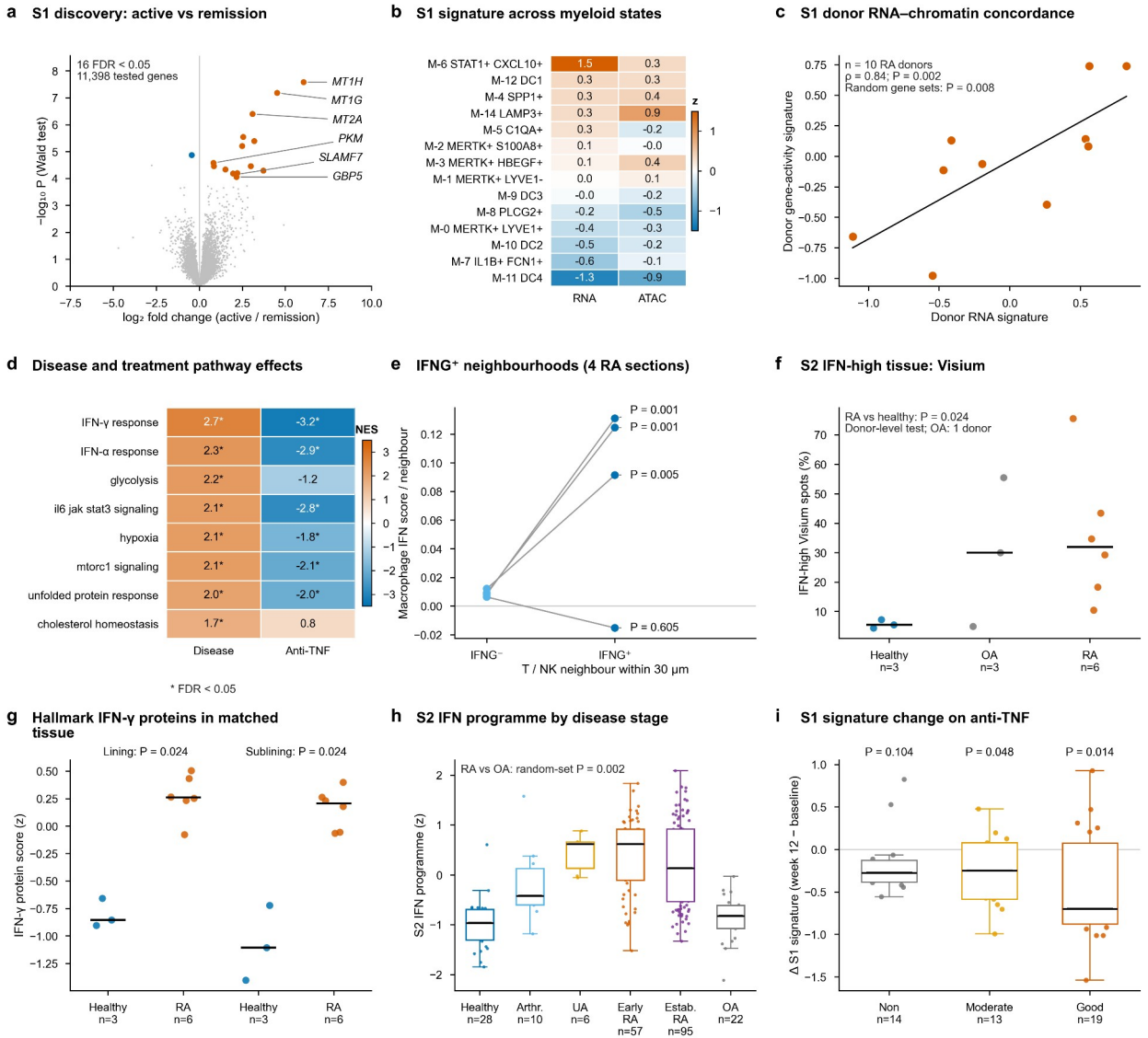
Separate discovery, tissue and treatment evidence

S1 maps to M-6. Its donor inflammation and bulk RA/OA validation tests are negative.

S2 shows spatial and bulk disease-stage associations.

Protein points and test use the same donor averages: 6 RA versus 3 healthy.

S1 falls in good responders, but the change is not significantly different from nonresponders.



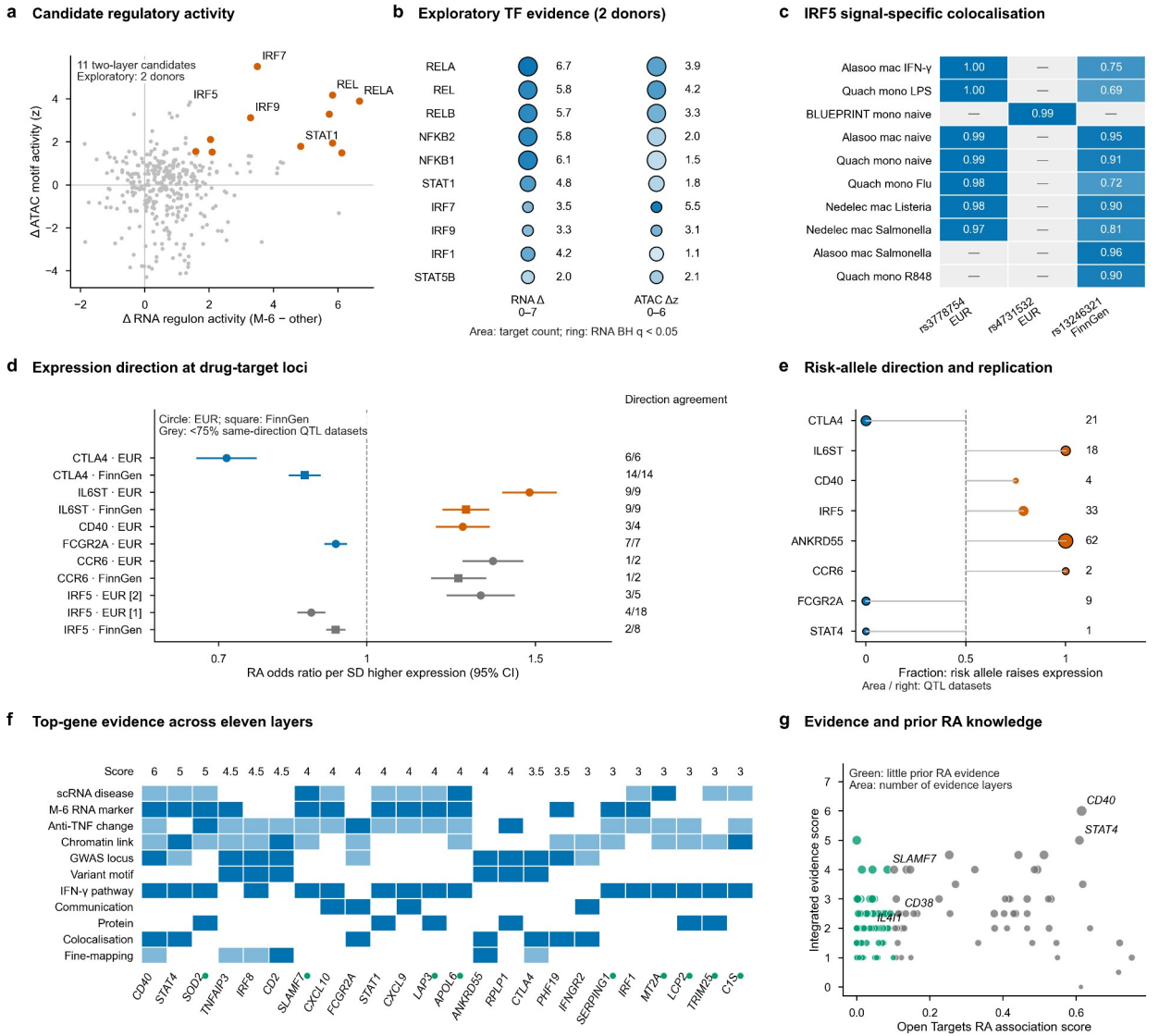
Candidate regulators and expression-associated genetic estimates

TF screen: two donors with M-6 and BH-adjusted gene-label permutations. Two-donor uncertainty remains.

IRF5 colocalisation supports a shared association signal.

Wald ratios cannot estimate drug efficacy or rule out pleiotropy.

Abatacept binds CD80/CD86. IL6ST is not IL6R, the target of tocilizumab.



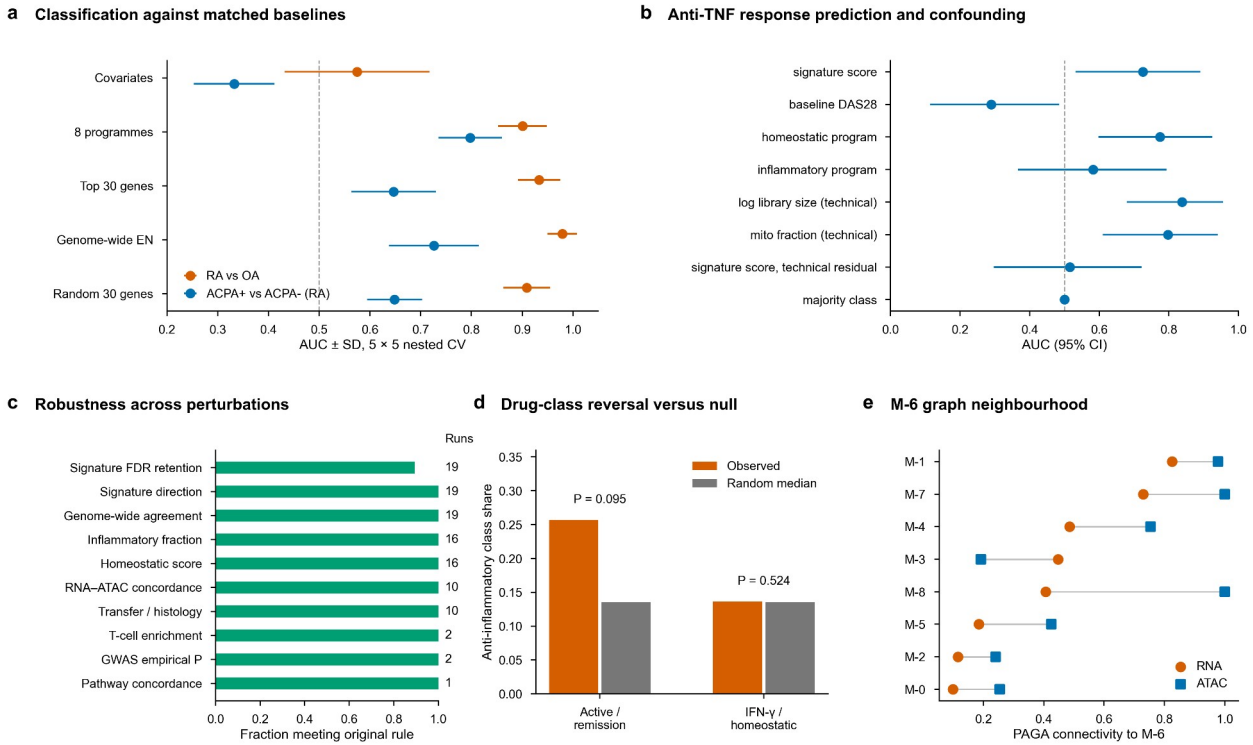
Baselines and boundary findings

Classifiers fit all preprocessing on training data only and use repeated nested CV.

199 label permutations repeat the same tuning and resampling as the observed model.

Random controls match the number of retained prioritised genes in each training fold.

Limits in blood transfer, response prediction, deconvolution and trajectory are all reported.



f Boundary findings and negative controls

Blood monocytes
Paired signature test: P = 0.45; 13 pairs

Bulk deconvolution
Failed the original sanity checks

Active / remission classification
AUC 0.50–0.60 in the original comparison

RA / OA: prioritised versus random genes
F2 vs random empirical p 0.2475

Bulk Tph score, adjusted for T-cell content
Random-set P = 0.81

Myeloid S-LDSC
Negative conditional τ z in both RA GWAS

Trajectory timing
Chromatin-before-RNA is not interpretable

East Asian T-cell peak enrichment
1.26×; P = 0.22

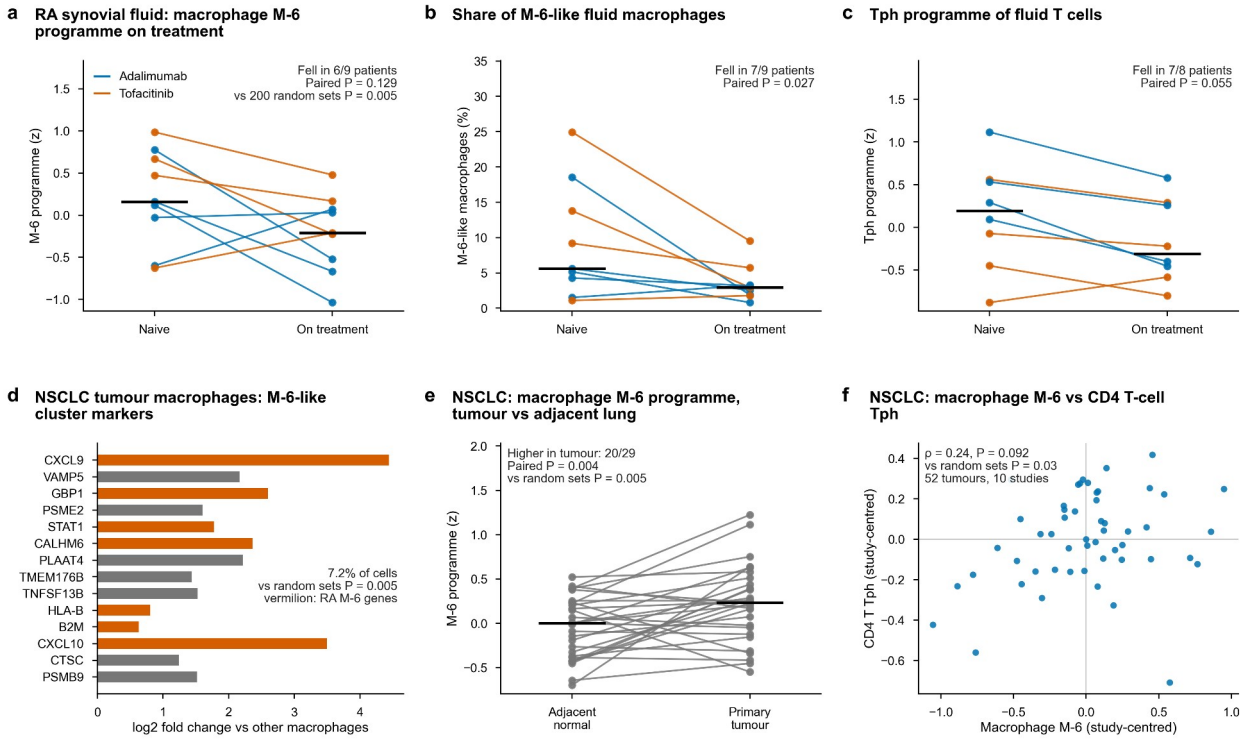
Exploratory transfer to synovial fluid and NSCLC

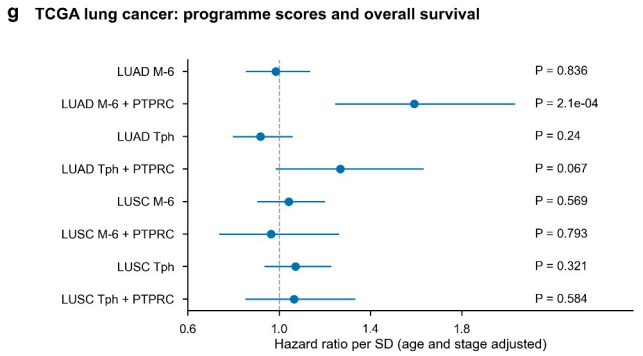
Fluid: 9 paired patients, two therapies pooled. No response labels are available.

The M-6-like fraction falls in 7/9 patients, but the paired pseudobulk-score test is not significant.

LuCA: the M-6 score is higher in tumour in 20/29 paired donors.

Tph coupling P = 0.092. The primary TCGA survival tests are negative.





h Pre-specified tests and verdicts

RA fluid, 9 patients (GSE296117)	
V1 M-6-like cluster exists	borderline (P = 0.055)
V2 M-6 programme falls on treatment	vs random P = 0.005; paired P = 0.129
V3 M-6-like share falls	supported (P = 0.027)
V4 Tph programme falls	borderline (P = 0.055)
NSCLC, LuCA atlas + TCGA	
N1 M-6-like tumour macrophages	supported (P = 0.005)
N2 higher in tumour than adjacent lung	supported (P = 0.004)
N3 couples with CD4 Tph	suggestive (ρ 0.24, P = 0.092)
N4 bulk survival	no association; LUAD HR 1.6 after PTPRC

Internal classifiers: final held-out performance

Mean AUC across 25 outer folds. All fitted transformations stay inside the training folds.
F0 covariates · F1 programmes · F2 prioritised genes · F3 genome-wide

RA vs OA

F0 0.574 ± 0.143

F1 0.900 ± 0.048

F2 0.933 ± 0.042

F3 0.979 ± 0.029

F1 permutation p 0.005

F2 vs random empirical p 0.2475

ACPA+ vs ACPA- (RA)

F0 0.332 ± 0.079

F1 0.797 ± 0.062

F2 0.647 ± 0.083

F3 0.726 ± 0.089

F1 permutation p 0.005

F2 vs random empirical p 0.5149

SD across correlated folds is not a confidence interval. No external clinical validation.

What the data support

A reproducible map of complementary RA associations and candidate targets for follow-up.

Score definitions, cohort independence and strength of evidence are kept separate.

Published states and original discoveries are clearly credited to their sources.

Causal regulation, therapeutic efficacy and clinical prediction need further validation.

Reproducibility and submission contents

The full pipeline was rerun from raw inputs in a separate project. Run logs and a comparison report come with the submission.

Saved copies of the API/GMT/eQTL inputs keep the reference version, including Open Targets 26.09.

A quick check verifies hashes, reruns three small analyses and draws six figures without the large raw/interim data.

README · Sources · Code · Processed tables · QC report · Limits · Presentation · Manifest

Sources and contact

Gupta et al., Nature Genetics 2023: GSE243917 atlas and author-defined states.

Alivernini et al., Nature Medicine 2020: E-MTAB-8322 discovery cohort.

GSE198520 / GSE89408: independent bulk cohorts.

GSE347958 / GSE347797 / PXD062696: shared spatial/protein cohort.

GSE296117 / LuCA / TCGA: exploratory extensions.

All accessions and licences: docs/sources.tsv

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